



DPUEnvControl

Using **DPUEnvControl**, weather conditions and the lighting state in the simulation environment can be specified.

Please also check the documentation for DPUEnvSCNX, which provides additional functionality compared to DPUEnvControl.

SILAB's environmental simulation is a two-step process:

- In a first step, the values required for graphical presentation are computed, based on generally understandable environmental *variables* (such as time of day, time of year, air humidity, and precipitation density). These computed values are in turn available as variables.
- In a second step, other DPUs (e.g., traffic simulation **DPUTRFX** and graphics engine **DPUSGE-World**) query the values of computed variables and display the resulting environment.

For each *variable*, a valid range (from minimum to maximum value) is defined, within which it can be modified. In order to smoothly modify changes to environmental conditions, a *ChangeRate* (in 1/s) is defined for each variable, which is used to perform the transitions.

This works as if DPUEnvControl contained a (DPU)Limiter for each *variable*, whose *MaxSlope* and *MinSlope* are set to *ChangeRate*.

These computations are grouped into *modules*. Each of the *modules* can optionally be deactivated. If so, the user himself needs to take care that the variables are set which would normally be computed by the module. Otherwise, they just remain at their last value or their default value.

A list of all *variables* and *modules* follows after the description of DPUEnvControl's DPU interface.

1. Inputs for each Variable:

<i>In</i> [<i>VarName</i>]	Input used to set the target value of variable [<i>VarName</i>]
<i>ChangeRate</i> [<i>VarName</i>]	Input to set the change rate of variable [<i>VarName</i>]

2. Dynamic parameters:

<i>ShowDialog</i>	Shows a dialog window (on the computer where DPUEnvControl is running) which can be used to change the values of all <i>Variables</i> .
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3. Outputs for each Variable:

<i>Out</i> [<i>VarName</i>]	Output which shows the current value of variable [<i>VarName</i>]
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4. Dynamic parameters for each Module:

Auto[ModuleName] Boolean value (0 or 1), which enables the automatic calculation of values of module *[ModuleName]*.

5. Dependencies to other DPUs:

There are no dependencies to other DPUs.

However, many other DPUs depend on the environment simulation. Thus, either *DPUEnvControl* or *DPUEnvSCNX* should be run on each computer which simulates traffic, pedestrians, or provides a graphical view.

6. Available Variables:

The following table lists all variables which cannot be computed automatically by a *Module* and thus should be set by the scenario developer.

Variable	Description	Default	Minimum	Maximum	Def.Change Rate
TimeOfDay	Time of day, in hours	12	0	24	0.1
TimeOfYear	Time of year, in months	6	0	12	0.1
Latitude	Geographical latitude, in degrees	49.833	-90	90	1.0
Humidity	Air humidity	0.13	0	1.5	0.02
HeadLight	Type of front light that is switched on	0	0	2	∞
LightingMode	Mode of environmental lighting	0	0	2	∞
ReflectionVisible	Is a reflective surface visible (allows optimization)	1	0	1	∞
Precipitation	Amount of precipitation	0	0	1	0.02
Thunderstorm	Intensity of thunderstorm (lighting and thunder)	0	0	1	0.02
Temperature	Temperature (in °C)	20	-100	100	1.0
NightBrightness	Intensity of ambient light at night	0.1	0	1	0.01

7. Available Modules:

Module ArtificialLight

The Module *ArtificialLight* determines a recommendation if artificial lights (such as vehicle head lamps and streetlights) should be turned on, based on the current lighting conditions. The darker it is, the higher the result (Variable *ArtificialLight*) will be. The threshold at which head lamps are actually turned on can be configured in the *DPUTRFX*.



Variable	Description	Minimum	Maximum	Def.Change Rate
ArtificialLight	Recommendation for turning on artificial lights	0	1	1.0

Module Light

Based on the current lighting conditions, the Module Light computes the intensity of the individual color channels of sun and ambient light. Additionally, the sun's position and a multiplier for the brightness of artificial light sources are computed.

Variable	Description	Minimum	Maximum	Def.Change Rate
AmbR	Red channel of the ambient light	0	1 ¹	1.0
AmbG	Green channel of the ambient light	0	1 [*]	1.0
AmbB	Blue channel of the ambient light	0	1 [*]	1.0
SunR	Red channel of the sun light	0	1 [*]	1.0
SunG	Green channel of the sun light	0	1 [*]	1.0
SunB	Blue channel of the sun light	0	1 [*]	1.0
SunPosX	X component of the direction in which the sun is located	-1	1	1.0
SunPosY	Y component of the direction in which the sun is located	-1	1	1.0
SunPosZ	Z component of the direction in which the sun is located	-1	1	1.0
Artificial Intensity	Multiplier for the brightness of artificial light sources	0	100	1.0

Module Fog

This module computes color and density of the fog. These depend on the current air humidity (Humidity).

Variable	Description	Minimum	Maximum	Def.Change Rate
FogR	Red channel of the fog color	0	1 [*]	1.0
FogG	Green channel of the fog color	0	1 [*]	1.0
FogB	Blue channel of the fog color	0	1 [*]	1.0
Fog	Fog density for use in OpenGL (exponential scale)	0	1 [*]	1.0

¹ Values above 1 can be set, but that is normally not useful.



Depending on the setting of Humidity, the following effects can be achieved:

Humidity	Effect
0 ... 0.1	Extremely clear vision (not realistic)
0.1 ... 0.25	Normal amount of atmospheric haze. The panorama is slightly hazy, but the 3D objects positioned in the landscape are clearly visible.
0.25 ... 0.5	Increased amount of atmospheric haze. The panorama is heavily hazy, but the landscape objects are still clearly visible.
0.5 ... 0.75	Light fog together with an overcast sky. Sky and panorama are no longer visible, distant landscape objects are hazy. A value of 0.75 corresponds to a range of sight of approx. 500 meters.
0.75 ... 1.0 ... 1.5	Dense fog. A value of 1.0 corresponds to a range of sight of approx. 50 meters.

8. Example:

The following example can be used to control the environmental behaviour on the status of the headlight selection from the mockup. Note that this is usually already included in the base configuration of SILAB (using DPUEnvSCNX).

```
DPUEnvControl Environment
{
    Computer = {SCN};
    Index = 42;
};
DPUMixAmplifier ~HLMux
{
    Computer = {MOCKUP};
    Index = 20;
    Multiplier2 = 2;
    MaxOut = 2;
};
Connections = {
    Mockup.LowerBeam -> ~HLMux.In1,
    Mockup.UpperBeam -> ~HLMux.In2,
    ~HLMux.Out -> Environment.InHeadLight
};
```